Registration No.: | | | | | | | | | | |

Total Number of Pages: 02 **Course: B.Tech, IDD (B.Tech and M.Tech)**
Sub_Code: REC3C001

3rd Semester Regular/Back Examination: 2023-24
SUBJECT: Analog Electronic Circuits
BRANCH(S): AEIE, BIOMED, EEE, ELECTRONICS & C.E, EIE, ELECTRICAL, ECE, ETC, EE, ECE
Time: 3 Hour
Max Marks: 100
Q.Code: N451
Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

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Part-I

Q1 **Answer the following questions:** **(2 x 10)**

a) What is the significant of pinch off voltage and threshold voltage in MOSFET?
b) Discuss about load line in a BJT biasing circuit.
c) Explain the roles of different external capacitors used in an amplifier circuit.
d) What is a super-beta transistor? Justify.
e) Whether the output of a CE configuration is in phase or out of phase with the input? Justify
f) Draw the hybrid and π model of a transistor in CE configuration. Write the difference between them.
g) With a neat sketch locate the parasitic capacitances for high frequency response. Among them which one is largest and which one is smallest?
h) Differentiate between small-signal amplifier and large-signal amplifier.
i) What are the advantages and disadvantages of negative feedback in an amplifier?
j) How the class A, B, C amplifiers are defined? Briefly discuss class A type.

Part-II

Q2 **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6 × 8)**

a) Explain how the transconductance of D-MOSFET is determined both graphically and mathematically?
b) Draw and explain the circuits of self-bias and Emitter bias. Write the differences between.
c) For a base-biased transistor circuit with V_{CC} = 30V, R_B = 300 KΩ, R_C = 2 KΩ, R_E = 1 KΩ and current gain = 100, find the Q-point, I_C(sat), V_C, and V_E.
d) What is source-degenerative resistance? Considering this resistance, draw the equivalent model of a CS MOSFET amplifier. Find its input impedance, output impedance, open loop voltage gain and the voltage gain in consideration of the load resistance.
e) Explain in detail the effect of R_S and R_L on BJT low frequency response.

f) What is modeling of a device? Draw the r_e models of all three configurations. Draw the hybrid model of a transistor in CE and derive the expressions for input impedance, output impedance, voltage and current gain.

g) Derive the expression of A_v and Z_o of the self-biased CS MOSFET circuit with un-bypassed R_S.

h) What is trans-conductance and overdrive voltage of a MOSFET? Write the relation between trans-conductance and overdrive voltage. Describe in detail, how the T equivalent model of a MOSFET is developed.

i) Draw and explain the small signal analysis of JFET CS and CD configuration.

j) What is a feedback network? Describe a positive feedback circuit as an oscillator. Explain Barkhausen's criteria for oscillation.

k) Explain the operation of an Op-amp as a non-inverting amplifier, a differentiator, and an Integrator.

l) Draw and explain the operation of an Instrumentation amplifier. What are the advantages of an Instrumentation amplifier?

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 With suitable diagrams, describe the operational principle of FETs and MOSFETs. Extend your discussion to V-I Characteristics of E- MOSFET and D-MOSFET. **(16)**

Q4 How many biasing circuits are there in a BJT circuit? Explain each of it. Out of all the biasing circuits which one is practically not employed? **(16)**

Q5 Which capacitors affect the low frequency and high frequency responses of a BJT amplifier circuit? What is Miller effect capacitance? Describe all in detail. **(16)**

Q6 What is a power amplifier circuit? Why is it named so? Explain class A, class B, class AB, and class C power amplifiers providing their efficiencies. **(16)**

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